CM3070 Final Project: Midterm Webinar Notes

Focus: Preliminary Report, Academic Writing, Design, Feature Prototype, Video

Based on the midterm webinar delivered by **Dr Sarah Rauchas**

Module:	CM3070 Computer Science Final Project
Session:	Midterm webinar (preliminary report focus)
Purpose:	Capture requirements, constraints, expectations, and Q&A guidance

December 14, 2025

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1 Webinar Slide Deck Screenshots

1.1 Slide 1: Title

Figure 1: Final Project: Mid-term webinar (Dr Sarah Rauchas).

1.2 Slide 2: Plan for today

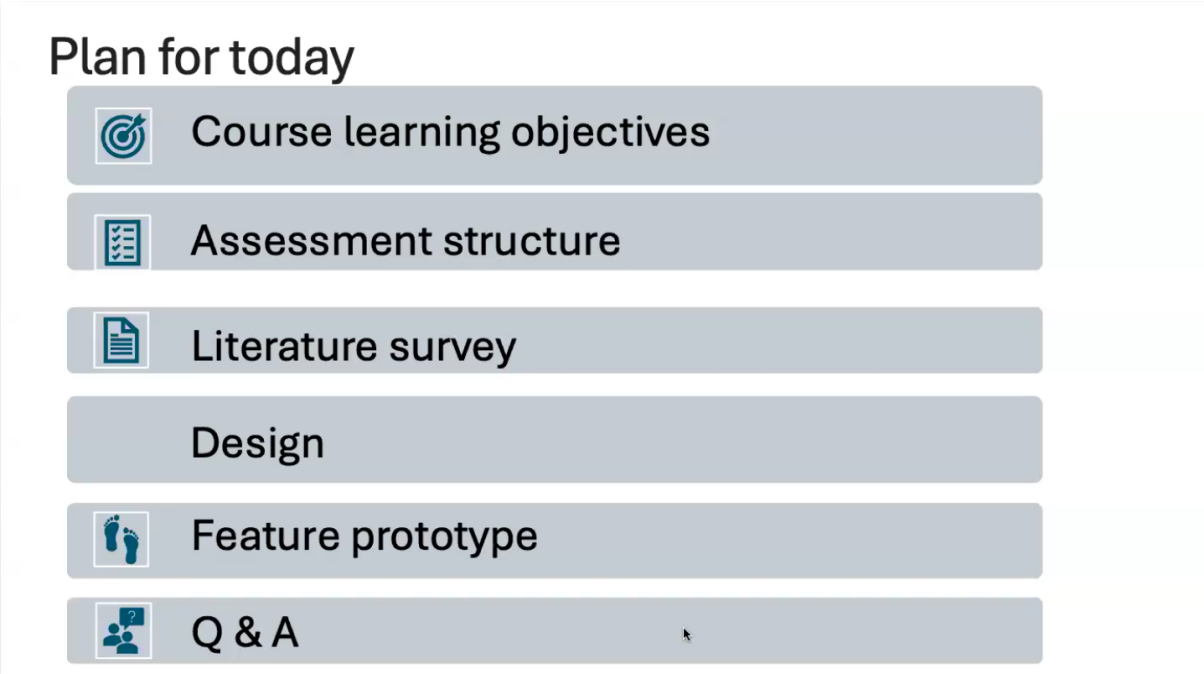

Figure 2: Agenda: learning objectives, assessment structure, literature survey, design, feature prototype, Q&A.

1.3 Slide 3: Course learning objectives

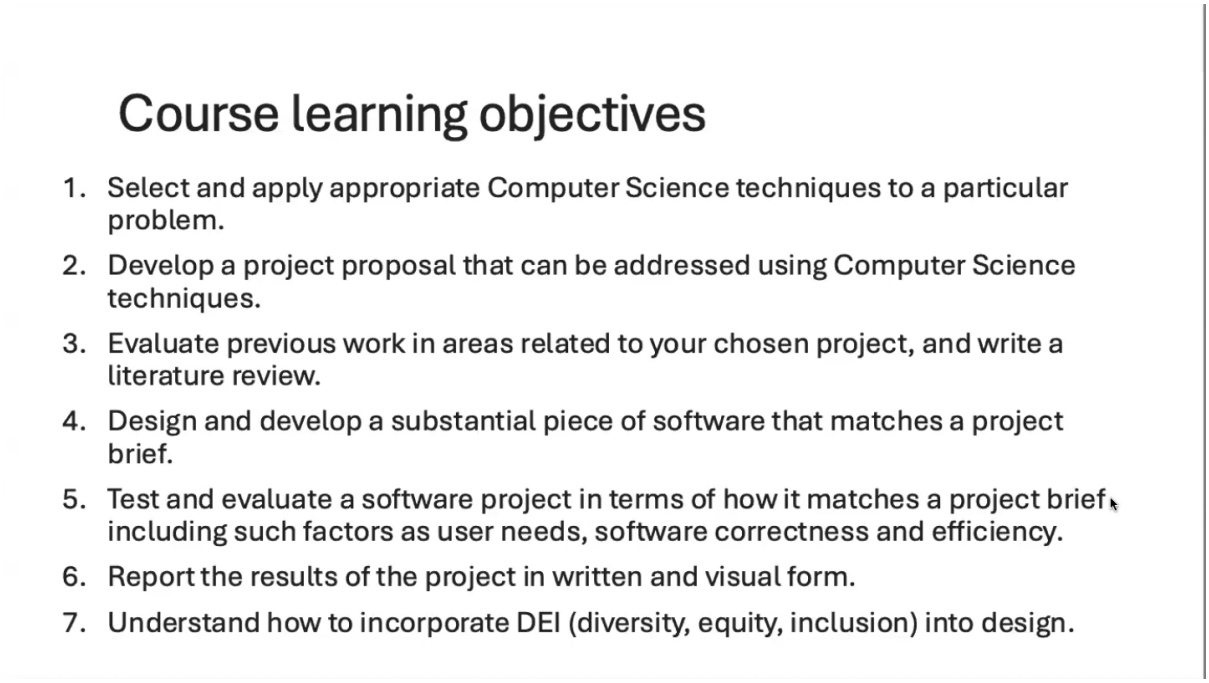

Figure 3: Course learning objectives for CM3070.

1.4 Slide 4: Assessment structure

Item	Deadline week		Weight (%)
Project proposal	4		0%
Preliminary project report	10		10%
Check-in quizzes	1-20		5%
Written exam	22		20%
Project presentation video	24		5%
Final project report and code	24		60%
Draft report	18		0%

Figure 4: Assessment items, deadline weeks, and weights.

1.5 Slide 5: Halfway through

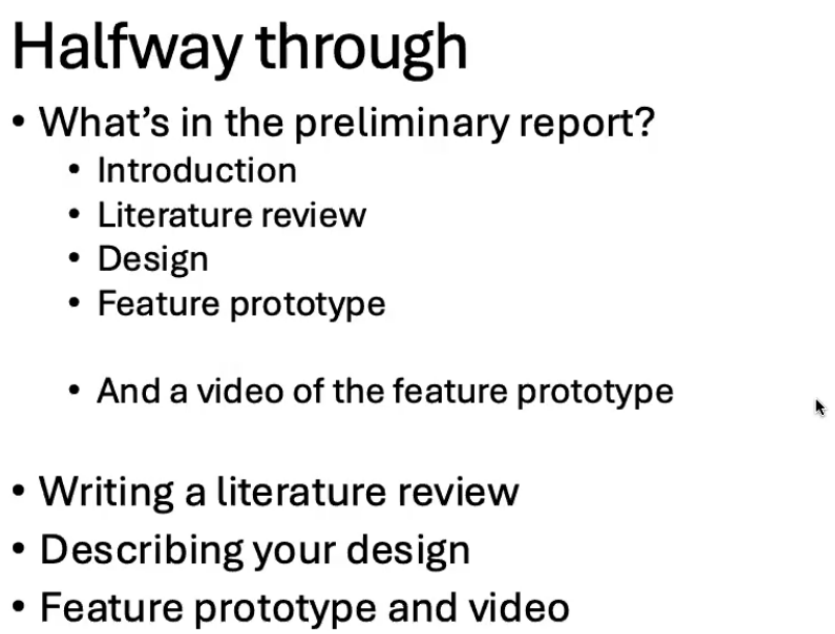

Figure 5: Preliminary report contents and today's focus.

1.6 Slide 6: Literature review and academic writing

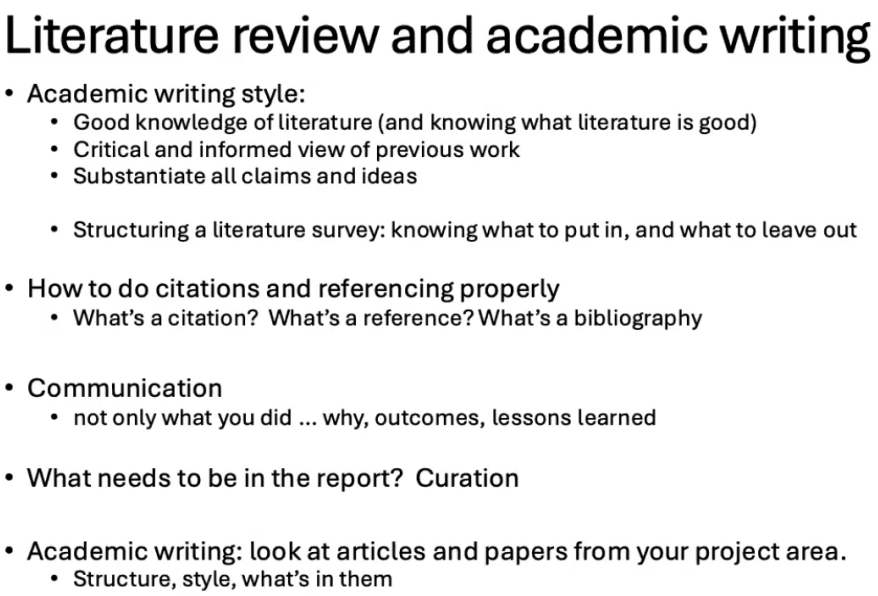

Figure 6: Academic writing expectations and literature review guidance.

1.7 Slide 7: Design

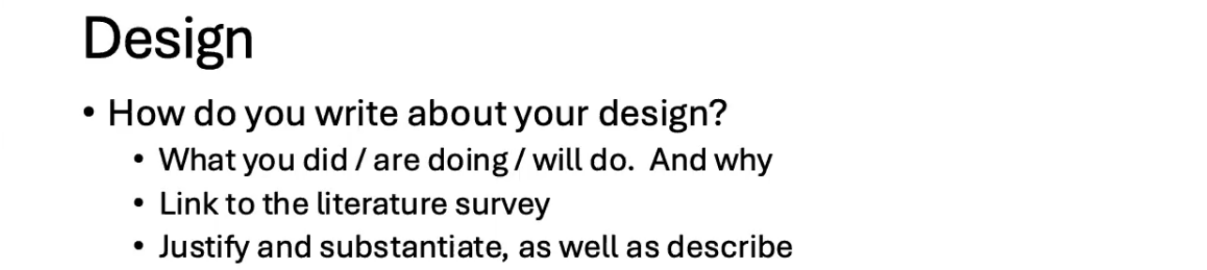

Figure 7: Design writing: what, why, link to literature, justify and substantiate.

1.8 Slide 8: Feature prototype

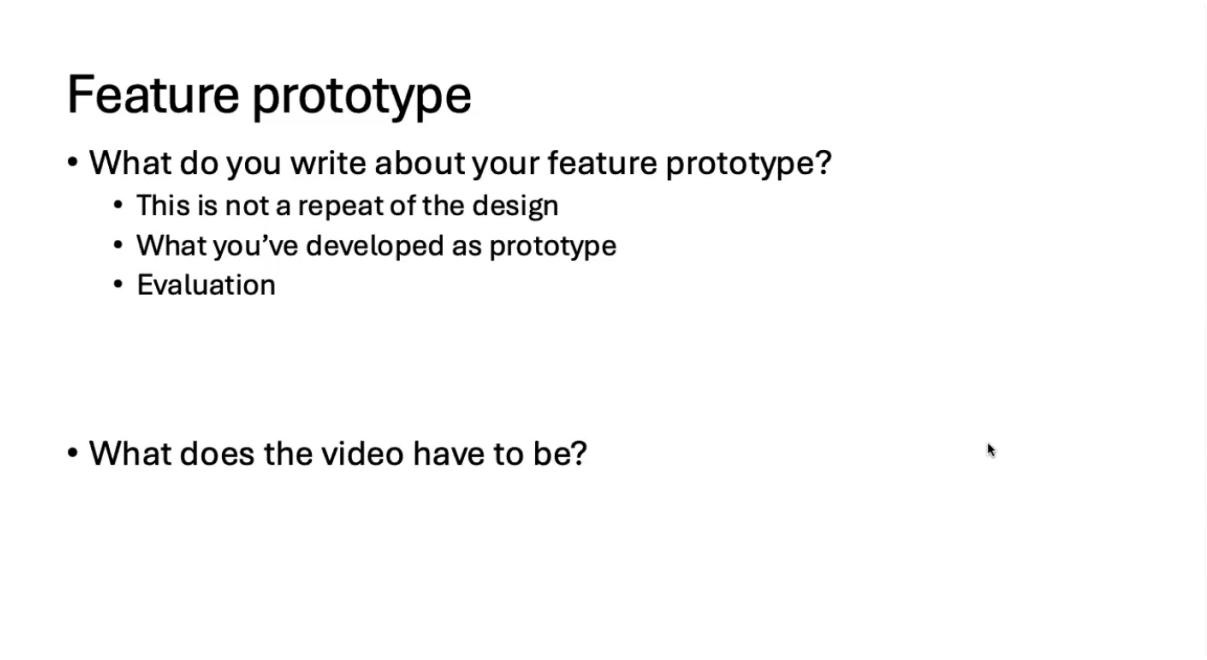

Figure 8: Feature prototype chapter: not a repeat of design, include evaluation, plus video requirements.

2 Webinar Context and Attendees

2.1 Session opening and locations

- Webinar begins at 10:00 in the UK, noting different time zones globally.
- Host mentions being in Johannesburg at the time of the session.
- Students are invited to share where they are joining from.

2.2 Tutors who introduced themselves

- **Fawad**: speaking from Malta; supervisor; PhD in Artificial Intelligence; supervising Group 18.
- **Colin**: speaking from Underberg, South Africa; looking forward to working with his group.
- **Ali**: speaking from Saudi Arabia; in charge of Group 20; long time module support.
- **Jordan (Yunus)**: speaking from Newcastle; PhD in Computer Science; managing Group 21.

2.3 Technical interruption

- A brief screen share interruption occurs.
- Tutors indicate slides are not visible, and the host re-shares the screen.

3 Webinar Agenda (Expanded Notes)

- Review course learning objectives.
- Review assessment structure and current position in the module timeline.
- Focus on upcoming submission needs:
  1. Literature review (literature survey) and academic writing expectations
  2. Design description
  3. Feature prototype chapter, including evaluation
  4. Feature prototype video requirements and strategy
- Q&A throughout (students type questions; tutors may respond in chat).

4 Course Learning Objectives (as shown on the slide)

1. Select and apply appropriate Computer Science techniques to a particular problem.
2. Develop a project proposal that can be addressed using Computer Science techniques.
3. Evaluate previous work in areas related to your chosen project, and write a literature review.
4. Design and develop a substantial piece of software that matches a project brief.

5. Test and evaluate a software project in terms of how it matches a project brief including user needs, correctness and efficiency.
6. Report the results of the project in written and visual form.
7. Understand how to incorporate DEI (diversity, equity, inclusion) into design.

4.1 Objectives emphasized for this stage

- Progress strongly on understanding context, prior work, and design.
- Development is not expected to be finished yet.
- Full testing and evaluation are not expected yet, except early prototype evaluation.
- Reporting of results is expected in limited form now, mainly from the feature prototype.
- DEI is expected throughout.

5 Assessment Structure and Deadline Rules

5.1 Assessment structure (from the slide)

Item	Deadline week	Weight (%)
Project proposal	4	0
Preliminary project report	10	10
Check-in quizzes	1 to 20	5
Written exam	22	20
Project presentation video	24	5
Final project report and code	24	60
Draft report	18	0

5.2 Strict deadline policy (critical)

- Some late submissions were accepted earlier due to Coursera platform glitches.
- Going forward, late submissions will generally not be accepted.
- You can submit early. You should not submit late.
- Exceptions are mainly platform-side issues or course-side issues that impact submission ability.
- Do not contact tutors one or two weeks after a deadline asking for late review of missed submissions.

5.3 Feedback and marking

- The preliminary report is a summative submission worth 10%.

- It is marked by at least two people, providing balanced feedback.
- This is intended to produce detailed, actionable guidance for the rest of the project.
- Feedback is cumulative across submissions, so fix issues early and carry improvements forward.

6 Preliminary Report: Required Components and Word Limits

6.1 What must be submitted

- A written report consisting of **four chapters**.
- A **feature prototype video**.
- Submit as a **PDF** (non-PDF is considered incorrect).

6.2 Four required chapters

1. Introduction
2. Literature review
3. Design
4. Feature prototype

6.3 Word limits

Section	**Maximum words**
Introduction	1000
Literature review	2500
Design	2000
Feature prototype	1500
Overall report total (excluding title page, references, etc.)	**6000**

6.4 Important note on word limits

- The per-chapter maximums are separate limits, but you must still stay within the overall 6000-word cap (excluding title page, references, etc.).
- Plan chapter lengths so the total remains within the overall limit even if individual chapter maxima would allow more.

6.5 Introduction compliance requirement (non-negotiable)

- Introduction must state which project template you are using.
- Include both the template number and the template title.
- Mention the template number and title in the video as well.

7 Academic Writing Expectations

7.1 Situate your report in context

- Your report should be grounded in an academic, commercial, or development context.
- Avoid writing that relies only on personal opinion or unsupported claims.

7.2 Source quality matters

- Blogs can be useful, but quality varies widely.
- Prefer credible background material as your core foundation.

7.3 Critical writing means balanced evaluation

- Critical does not mean negative.
- Critique includes strengths, limitations, tradeoffs, and assumptions.

7.4 Substantiate and justify

- Do not only describe. Explain the reasons and evidence.
- A useful mental model for the reader's questions:
  - What did you do?
  - Why did you do it?
  - What happened (outcome so far)?
  - What did you learn (takeaways)?

7.5 Avoid tutorial style

- Assume the reader understands general computer science.
- Do not write a textbook survey of basic theory.
- Focus on how methods have been applied in your project area and why you chose them.

8 Citations, References, Quotations, Bibliography

8.1 Definitions

- **Citation**: the in-text pointer to a source (author-year or numeric pointer).
- **Reference**: the full entry in the reference list matching the citation.
- **Quotation**: direct text copied from a source, requiring quotation marks and a citation; add page number where possible.
- **Bibliography**: optional list of reading not explicitly cited.

8.2 Citation and reference must match

- Every citation must map to an entry in the reference list.
- A single reference can be cited multiple times in different report sections.

8.3 Referencing style

- Any established referencing style is acceptable if used consistently.
- Do not invent your own referencing style.
- Practical approach: follow the style used by papers in your project area.

9 How to Write a Strong Literature Review

9.1 Purpose

- Set the context for the rest of the report.
- Explain what exists, what is relevant, and what gaps or opportunities your project targets.
- Help the reader understand why you chose approach A rather than approach B.

9.2 Minimum evidence baseline (practical)

- Aim for at least 4 to 6 credible sources as a baseline in the literature review, then add more only if they improve synthesis and relevance.

9.3 Do not just summarize

- A strong literature review weaves sources together into a narrative specific to your project.
- Compare and contrast. Show relevance. Highlight differences in viewpoint or evidence.
- The marker should see your synthesis, not just a list of paper summaries.

9.4 Include market solutions if appropriate

- If your project has real market equivalents, discuss available solutions.
- Also include literature behind those solutions or closely related research where possible.

9.5 Curation model (exhibition analogy)

- You choose what belongs in the report based on what the reader needs to know.
- Include content that supports your project theme and answers likely reader questions.

9.6 Recommended practice

- Read papers in your project area and observe structure, style, and what they include.
- Notice that good papers do not provide long tutorials of basic theory.

10 Design Chapter Guidance

10.1 What to write about

- Explain what you did, are doing, and will do.
- Explain why you chose this approach.
- Justify and substantiate design decisions using evidence.

10.2 Link design to literature and market findings

- Design should not be isolated from the literature review.
- Explicitly connect design choices to what you found in research and market analysis.

10.3 Prototypes can change direction

- If evidence shows a chosen approach is not working, changing course is acceptable.
- It is not framed as failure if you show reasoning, iteration, and learning.

11 Feature Prototype Chapter: What is Required

11.1 Not a repeat of design

- Feature prototype chapter is not a second design chapter.
- It should describe what you actually implemented so far, plus evaluation and lessons learned.

11.2 Backend-only prototype is acceptable

- If you built a backend first, demonstrate it using:
  - a test harness
  - scripted inputs and outputs
  - logs or a minimal interface

11.3 Evaluation is required now

- There are marks for evaluation within the preliminary report.
- Start evaluation now on what you have built so far.
- Testing and evaluation are different:

- **Testing**: correctness, does it run, does it do what it claims.
- **Evaluation**: effectiveness against goals, quality of outcomes, comparisons, metrics.

12 Video Requirements and Strategy

12.1 Strict constraints

- Video must be between **3 and 5 minutes**.
- Normal speed narration. Do not speed up audio to squeeze in extra content.
- Must demonstrate the prototype. Slides alone are not sufficient.
- Use a normal human narration. Do not use synthetic or AI generated voice.
- Keep within the minimum and maximum duration. Do not speed up audio to fit more content.

12.2 What the video should show

- Prototype demonstration working or partly working.
- Motivation and relevance.
- Connection to established work (literature or market context).
- What you implemented and why it matters.
- Evaluation performed so far and what you learned.

12.3 If something breaks

- If it is not working, show it and explain why.
- A genuine work in progress can still be good if it reflects real iteration and effort.
- Lack of progress due to last minute work is viewed negatively.

12.4 Presentation choices affect grading

- You are graded on what you choose to include and exclude.
- Avoid irrelevant filler, reading slides verbatim, and chaotic screen recordings.
- Reduce friction for the viewer so the key achievements are obvious.

13 GitHub and Reproducibility Guidance

13.1 Repository expectations

- Host work on GitHub where possible.
- Include a GitHub link in the preliminary report if appropriate.

- Repository may need to be public during marking to reduce friction for markers.

13.2 Why GitHub is recommended

- Avoid uploading large zip files to the submission interface.
- Encourages professional practice with version control and reproducibility.

14 Ambition, Realism, and Results Interpretation

- Ambition can help, but must be balanced with realism.
- If you compare several models and results differ from expectations, it can still be good work if you:
  - explain the method clearly
  - justify decisions using evidence
  - analyze why outcomes differed

15 Tutor Support: How to Ask Effectively

- Tutors can support targeted concerns.
- Avoid asking for full report review without specific questions.
- Better approach:
  - Identify what you think is strong.
  - Identify what you are unsure about.
  - Ask a focused question about that uncertainty.

16 Q&A Highlights Captured

16.1 Prototype without UI

Backend-first is acceptable. Demonstrate via harnesses, scripts, and controlled examples.

16.2 Market solutions in background research

Include market solutions where relevant, and include literature alongside, not only product descriptions.

16.3 Open-source models and tools

Generally acceptable if you clearly explain what you used and how markers can access or reproduce it.

16.4 Front-end heavy caution

If progress is mostly generic UI and login flows, it can appear less substantive. Prioritise core technical substance and explain it clearly.

16.5 Literature review size

Quality is not about number of papers. Prioritise relevance, synthesis, and credible sources.

17 Action Checklist (Preliminary Report Submission)

17.1 Compliance checklist

1. Submit a PDF.
2. Exactly four chapters: Introduction, Literature Review, Design, Feature Prototype.
3. Keep within the overall 6000 word limit and the chapter maxima.
4. Introduction states template number and template title.
5. Citations and references match; consistent referencing style throughout.
6. Literature review synthesises and critiques, not a tutorial or paper-by-paper summary.
7. Design chapter links decisions to literature and market insights.
8. Prototype chapter describes what was built and includes evaluation so far.
9. Video is 3 to 5 minutes, normal speed, and demonstrates the prototype.
10. GitHub link included when helpful to reduce friction for marking.

17.2 Marks-focused checklist

1. Claims are substantiated with evidence.
2. Reader questions are anticipated: what, why, outcome, takeaways.
3. Evaluation is framed as effectiveness, not only correctness.
4. Presentation choices are tight and relevant.
5. Ambition is balanced with realistic delivery and justified tradeoffs.

18 Closing Remarks Captured

- Encouragement: do the work, do good work, and enjoy the work.
- Tutors contributed useful advice in chat and wished students good luck.